CS3003: IT Project Management

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Part 5: Project Risk Management

Project Risk Management Overview (ISO 21500)

Subject groups	Process groups				
	Initiating	Planning	Implementing	Controlling	Closing
Integration	4.3.2 Develop project charter	4.3.3 Develop project plans	4.3.4 Direct project work	4.3.5 Control project work 4.3.6 Control changes	4.3.7 Close project phase or project 4.3.8 Collect lessons learned
Stakeholder	4.3.9 Identify stakeholders		4.3.10 Manage stakeholders		
Scope		4.3.11 Define scope 4.3.12 Create work breakdown structure 4.3.13 Define activities		4.3.14 Control scope	
Resource	4.3.15 Establish project team	4.3.16 Estimate resources 4.3.17 Define project organization	4.3.18 Develop project team	4.3.19 Control resources 4.3.20 Manage project team	
Time		4.3.21 Sequence activities 4.3.22 Estimate activity durations 4.3.23 Develop schedule		4.3.24 Control schedule	
Cost		4.3.25 Estimate costs 4.3.26 Develop budget		4.3.27 Control costs	
Risk		4.3.28 Identify risks 4.3.29 Assess risks	4.3.30 Treat risks	4.3.31 Control risks	
Quality		4.3.32 Plan quality	4.3.33 Perform quality assurance	4.3.34 Perform quality control	
Procurement		4.3.35 Plan procurements	4.3.36 Select suppliers	4.3.37 Administer procurements	
Communication		4.3.38 Plan communications	4.3.39 Distribute information	4.3.40 Manage communications	

NOTE The purpose of this table is not to specify a chronological order for carrying out the activities. Its purpose is to map subject groups and process groups.

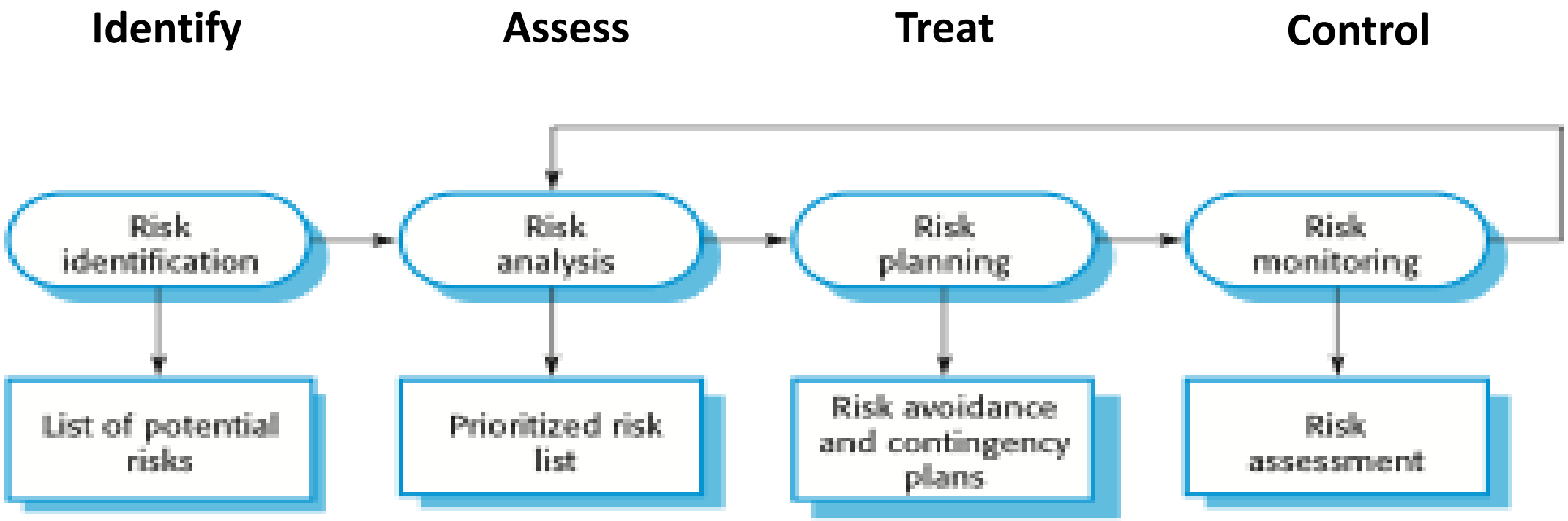
Project Risk Management

- **What is a Risk?** **“**An uncertain event or condition that, if it occurs, has a positive or a negative effect on a project’s objective” or “uncertainty of outcome” (Prince2)
- **Characteristics of a Risk**
  - A definable event
  - *Probability* of occurrence
  - *Impact* (consequence) of occurrence
- A degree of risk-taking is inevitable in projects. We will mainly focus on risks with negative effects.
- **What is Risk Management? “**Manage a project’s exposure to risk (that is, the probability of specific risks occurring and the potential impact if they did occur). The aim is to manage that exposure by taking action to keep exposure to an acceptable level in a cost-effective way.”

Project Risk Management

- **What is Risk Tolerance or Risk Appetite?** The amount of risk a project is prepared to tolerate (typically considered by PM, Project Sponsor and SC, taking into account the overall organization's risk tolerance).
- **What is a Risk Management Plan?** Documents the procedures for managing risks throughout the project, i.e. how risk management will be performed on the project. Part of the overall Project Management Plan.
- Main topics addressed in a Risk Management Plan:
  1. Methodology
  2. Roles and Responsibilities
  3. Budget and Schedule
  4. Risk Categories
  5. Risk Probability and Impact
  6. Risk Tolerances
  7. Risk Tracking and Documentation

Project Risk Management: The (generic) Risk Management Process



Project Risk Management: The (generic) Risk Management Process

- **Risk Identification**
  - Identify Threats (negative impact) or opportunities (positive impact). A repeatable process as a risk may change or new risks may emerge during the project.
  - Can be done via brainstorming, interviewing, workshops or individually (e.g. PM), checklists, previous projects, Risk Breakdown Structures.
  - Involved multiple participants, e.g. PM, Project Sponsor, SC, Project Team, senior managers, users, risk management experts, subject matter experts.
  - Risks should be specific, e.g. of the form: If <event> when <point in project> then <result>
  - The primary output are relevant entries in a **Risk Register** (or else known as **Risk Log** (Prince2))

Project Risk Management: The (generic) Risk Management Process

- **Sample Risk Register** (generic)

TABLE 11-5 Sample risk register

No.	Rank	Risk	Description	Category	Root Cause	Triggers	Potential Responses	Risk Owner	Probability	Impact	Status
R44	1										
R21	2										
R7	3										

from Schwalbe's book
.

Project Risk Management: The (generic) Risk Management Process

- **Risk Register/Log Elements** (Prince2)

Risk identifier: unique code to allow grouping of all information on this risk

Description: brief description of the risk

Risk category (e.g. commercial, legal, technical)

Impact: effect on the project/programme/organisation if this risk were to occur

Probability: estimate of the likelihood of the risk occurring

Proximity: how close in time is the risk likely to occur

Countermeasure(s): what actions have been taken/will be taken to counter this risk

Owner: who has been appointed to keep an eye on this risk

Author: who submitted the risk

Date identified: when was the risk first identified

Date of last update: when was the status of this risk last checked

Current status: e.g. dead, reducing, increasing, no change.

Project Risk Management: The (generic) Risk Management Process

- **Examples of Risk Categories and Risks** (not a full list to be used for reference purposes)

Risk Category	Example Risks
Technology	The database used in the system cannot process as many transactions per second as expected.
People	It is impossible to recruit staff with the skills required. Key staff are ill and unavailable at critical times. Required training for staff is not available.
Organizational	The organization is restructured so that different management are responsible for the project. Organizational financial problems force reductions in the project budget.
Tools	The code generated by software code generation tools is inefficient. Software tools cannot work together in an integrated way.
Requirements	Changes to requirements that require major design rework are proposed. Customers fail to understand the impact of requirements changes.
Estimation	The time required to develop the software is underestimated. The rate of defect repair is underestimated. The size of the software is underestimated.

Project Risk Management: The (generic) Risk Management Process

- **Assess (Risk Analysis):** measure and prioritize the risks for further action.
  - This process includes estimating the probability of occurrence of each risk and the corresponding impact for project objectives, if the risk does occur.
  - The risks are then prioritized in accordance with this assessment considering other factors such as the timeframe and key stakeholders' risk tolerance.
  - Impact is often considered in terms of time, quality, cost/benefit, people/resources.
  - Impact scale can be H/M/L or catastrophic, serious, tolerable or insignificant (or other similar scales)
  - Risks may be quantified (using a number of methods)

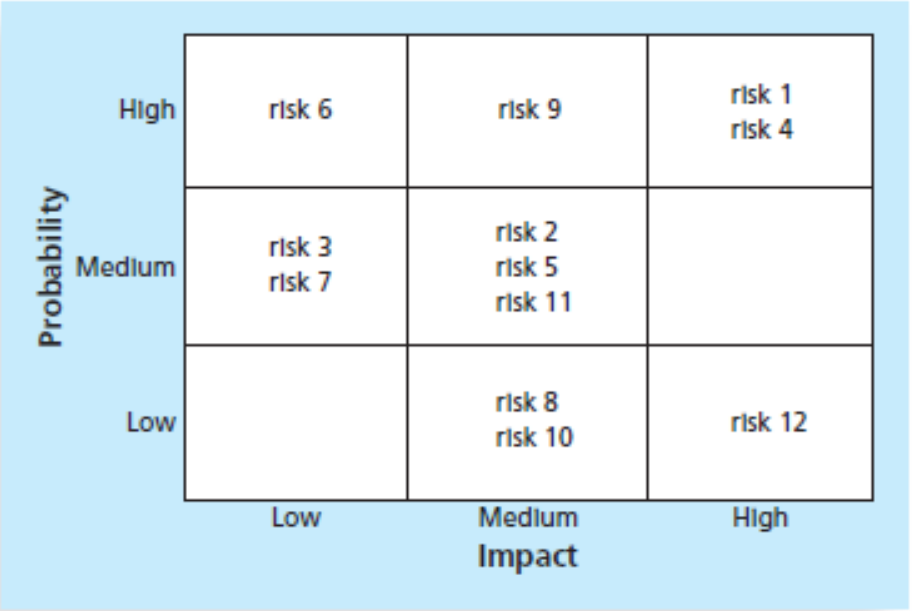


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FIGURE 11-5 Sample probability/impact matrix

Image from Schwalbe's book .

Project Risk Management: The (generic) Risk Management Process

- **Treat (Planning):** develop options and determine actions to reduce threats to project objectives. Actions typically break into the following 5 types (from Prince2)

Table 17.1

Prevention	Terminate the risk – by doing things differently and thus removing the risk, where it is feasible to do so. Countermeasures are put in place that either stop the threat or problem from occurring or prevent it having any impact on the project or business
Reduction	Treat the risk – take action to control it in some way where the actions either reduce the likelihood of the risk developing or limit the impact on the project to acceptable levels
Transference	This is a specialist form of risk reduction where the management of the risk is passed to a third party via, for instance, an insurance policy or penalty clause, such that the impact of the risk is no longer an issue for the health of the project. Not all risks can be transferred in this way
Acceptance	Tolerate the risk – perhaps because nothing can be done at a reasonable cost to mitigate it or the likelihood and impact of the risk occurring are at an acceptable level
Contingency	These are actions planned and organised to come into force as and when the risk occurs

Note. Treating risks implies including resources and activities into the project's budget and schedule. Risk treatment should be appropriate to the risk, cost-effective, timely, realistic within the project context, understood by all parties involved and assigned to an appropriate person.

Project Risk Management: The (generic) Risk Management Process

- **Control (Monitoring)**
  - Purpose is to minimize disruption to the project by determining whether the risk responses are executed and whether they have the desired effect.
  - It is achieved by tracking the identified risks, identifying and analysing new risks, monitoring trigger conditions for contingency plans and reviewing progress on risk treatments while evaluating their effectiveness.
  - Project risks should be periodically evaluated throughout the project life cycle, when a new risk arises or when a milestone is reached.

Project Risk Management: The (generic) Risk Management Process

Main Responsibilities

Project Manager

- Responsible for the overall management of risk (a key-responsibility), including identifying risks herself. The PM should “eat, sleep and breath” with potential risks and risk management. Apart from considering uncertain events or conditions, key questions also include: “What we have not been doing, and we should have been doing (or plan to do)?” and “What are possible consequences of what we have been or not been doing?”
- Ensures that risks are identified, recorded and regularly reviewed.
- Modify project plan to include risk management.
- Timely communication is a critical success factor!

Project Teams

- Proactively identify risks and timely alert the Project Manager. Actively participate in treatment actions.

Project Steering Committee and/or Project Sponsor

- Notify PM of any external risk exposure
- Make decisions on recommended by the PM actions to risk
- Balance risk-taking and benefits of the project
- Notify programme management and/or organization of any risks that affect project’s ability to meet corporate or programme objectives.

Project Risk Management vis-a-vi ISO 21500 Process Groups

Planning

Identify Risks

Assess Risks

Implementing

Treat Risks

Controlling

Control Risks